TEXORA · LECTURE SHEET

Circular Single-Jersey Machine

One cylinder of latch needles, one set of sinkers, and as many feeders as the frame will carry: the single-jersey (S/J) circular machine is the volume engine of the knit industry — the T-shirt machine — and the reference against which rib, interlock and fleece machines are understood.

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1. Construction and yarn passage

Cones stand on an overhead or side **creel**. Each yarn passes through a tensioner and stop motion to a **positive/storage feeder**, then down to its **feeder guide** at the cylinder. The **cylinder** — a steel ring with vertical tricks (slots) carrying latch needles, typically 12–40 inch diameter, E14–E42 — either rotates past stationary cam boxes or stands still while the cam ring and feeders rotate around it; the knitting action is identical. Around the top of the cylinder sits the **sinker ring** with horizontal sinkers moving radially in and out by their own cam ring, timed against the needles. Inside the cylinder the formed tube is drawn down by **take-down rollers**, spread by a former, and wound into a roll beneath the machine. Drives, feeder count and diameter set the speed factor; a 30-inch machine at 25 rpm runs a cylinder surface speed of $\pi \times 0.762\ \text{m} \times 25 \approx 59.9$ m/min.

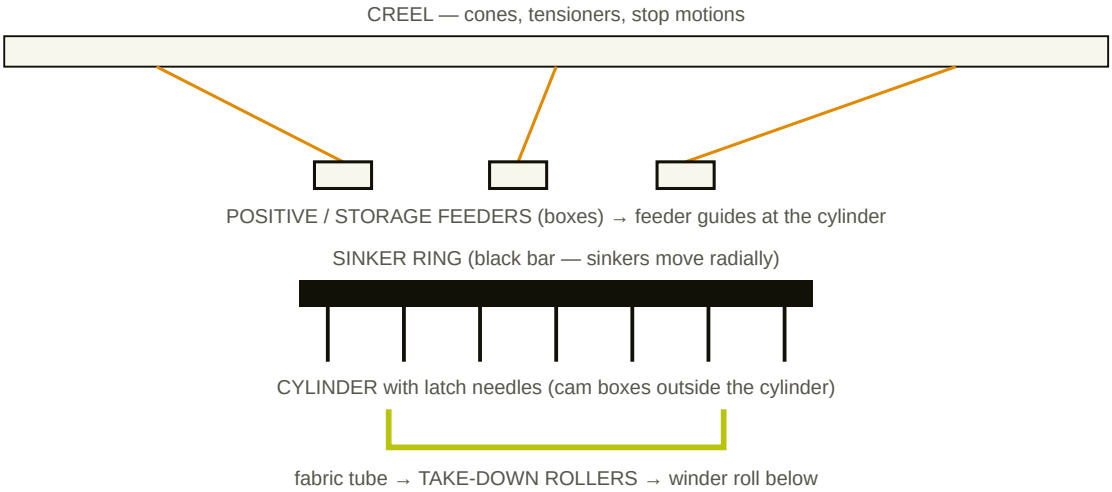


Figure 1 — Single-jersey circular machine, schematic section. Yarn: creel → feeder → needle cylinder; fabric: tube down through take-down to the roll. Either cylinder or cam/feed assembly rotates — the relative motion is what knits.

2. What each feeder does; stripers and attachments

Each feeder is a complete knitting station (Topic 3 cam set). With F feeders the machine knits F courses per revolution. Feeders may all knit the same yarn (plain fabric at full speed) or carry different colours/yarns: horizontal stripes result when consecutive feeders carry different colours — stripe length in courses equals the feeder repeat. Attachments extend the base machine: **auto-stripers** (finger selection of up to 4–6 yarns per feeder for engineered stripes), **plating feeders** (two yarns to the same hook in fixed face/back order), **fleece machines** (extra feeders laying a fleecing yarn held by sinker loops plus a binding yarn), **terry sinkers** (two sinker nib heights giving pile loops on one face), and elastane (lycra) feeders with their own positive feed for stretch jersey.

3. Production — worked example (verified)

WORKED EXAMPLE — S/J PRODUCTION IN KG/H AND M/H

Machine: D = 30 in, E24, F = 72 feeders, 25 rpm, stitch length l = 5.0 mm, yarn 30 tex, efficiency 85 %. Fabric CPC after relaxation = 11 courses/cm (the relaxed value belonging to this stitch length and yarn — see the consistency rule in Topic 12).

Needles N = π × 30 × 24 = **2262**. Yarn per course = 2262 × 5.0 mm = 11.31 m.

Courses/min = 72 × 25 = 1800. Yarn rate = 1800 × 11.31 = 20,358 m/min = 1,221,480 m/h.

Mass = 1,221,480 m × 30 g/km ÷ 1000 → 1,221.48 km × 30 g/km = 36.64 kg/h theoretical; × 0.85 = **31.1 kg/h actual**.

Length: courses/hour = 1800 × 60 = 108,000; ÷ 11 courses/cm = 9818 cm/h = 98.2 m/h theoretical; × 0.85 = **83.5 m/h**. (Check: 31.1 kg/h ÷ 83.5 m/h = 0.372 kg per metre of tube — the figure an inspector weighs against.)

Note what drives what: kg/h rises with feeder count, rpm, needle count and stitch length; m/h is fixed by courses/hour and CPC alone — lengthening the stitch raises kg/h but, by lowering CPC after relaxation, also raises m/h less than proportionally. Topic 12 formalises all of these.

4. Fabric quality on the S/J machine — the standing faults

- **Barré** — horizontal bands from feeder-to-feeder stitch-length or yarn differences; prevented by positive feed calibration and even yarn tension.
- **Needle lines** — vertical marks from bent/damaged needles or dirty tricks; cured only by needle/trick attention.
- **Holes and press-offs** — clearing, latch or yarn faults (Topic 3); stop motions must catch yarn breaks before a press-off runs.
- **Spirality** — wales skewing after washing, from yarn twist liveliness in an unbalanced structure; managed by twist choice, heat-setting and finishing, and measured as skew %.

- **Curling and width variation** — inherent to jersey (Topic 2); controlled by stitch length and finishing, and by cutting the tube on a marked wale.

5. Typical specification and product range

ITEM	TYPICAL S/J RANGE
Diameter	12–40 in (body-size 12–22 in seamless-ish tubes; open-width frames above)
Gauge	E14–E42 (E24–E28 the T-shirt workhorse)
Feeders	≈ 2.5–4 per inch of diameter (30 in → 72–108)
Speed	18–35 rpm; speed factor ≈ 0.7–1.1 m·rpm units as D(m)×rpm ≈ 14–27
Products	T-shirts, vests, dress jersey, fleece, terry, lacoste/pique derivatives

Glossary

Creel — cone stand supplying the machine. **Sinker ring** — ring carrying the sinkers around the cylinder top. **Speed factor** — D(m) × rpm, a size-corrected speed index. **Barré** — visible horizontal banding across courses. **Spirality** — skewing of wales relative to courses after relaxation/washing. **Plating** — two yarns knitted into the same loops in fixed order.

Key points

• S/J = one cylinder, sinkers, F feeders → F courses per revolution, fabric as a tube.
• Verified output for 30 in/E24/72F/25 rpm/5.0 mm/30 tex at 85 %: 31.1 kg/h and 83.5 m/h at relaxed CPC 11/cm.
• Quality is feeder uniformity (barré), needle condition (lines, holes) and stitch-length control; spirality and curl are structural, managed not eliminated.

Practice

1. Recompute the worked example at 30 rpm instead of 25 rpm (all else equal): kg/h and m/h at 85 % efficiency.
2. A 30-inch E24 machine must produce a tube that opens to 150 cm finished width. Using WPC after relaxation, what WPC is required? (Open width = half the tube circumference in wales.)
3. Explain why adding feeders raises kg/h but does not by itself change GSM.
4. Two adjacent fabric rolls differ: one shows barré, one shows needle lines. State the different origins and first checks for each.

5. Why is elastane always fed by its own positive feeder rather than through a storage feeder?

Answer key

1. Courses/min = 72 × 30 = 2160; yarn rate = 2160 × 11.31 = 24,430 m/min = 1,465,800 m/h; mass = 1465.8 × 30 g = 43.97 kg/h; × 0.85 = **37.4 kg/h**. Length: 2160 × 60 = 129,600 courses/h ÷ 11 courses/cm = 11,782 cm = 117.8 m/h; × 0.85 = **100.1 m/h**.
2. Tube circumference in wales = N = 2262 wales around; open width uses all 2262 wales across 150 cm → WPC = 2262 ÷ 150 = **15.1 wales/cm** required after relaxation.
3. GSM is set by stitch length, yarn tex and the resulting CPC × WPC. More feeders make more courses per minute of the same structure — more area per hour at unchanged loops/cm² — so kg/h and m/h rise together, GSM constant.
4. Barré: course-wise (horizontal) — check feeder-wise stitch length/positive-feed settings and yarn lot/tension differences between feeders. Needle lines: wale-wise (vertical) — inspect individual needles and tricks at the marked wales for bent latches, burrs and dirt.
5. Elastane must be delivered at a precisely controlled draft (stretch) so its retractive force — hence fabric stretch, width and GSM — is uniform; only a positive feeder meters length accurately enough. A storage feeder would let draft vary with package and tension.
